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Assignment-2

①

```
i) #include <stdio.h>
#include <stdlib.h>
int main() {
    FILE * fptr;
    fptr = fopen("text.txt", "w");
    fclose(fptr);
    return 0;
}
```

```
ii) #include <stdio.h>
#include <stdlib.h>
int main() {
    FILE * fptr;
    fptr = fopen("file.txt", "r");
    char data[100];
    if (fptr == NULL) printf("failed to open file.txt");
    else
        while (fgets(data, 100, fptr) != NULL) {
            printf("%s", data);
        }
    fclose(fptr);
    return 0;
}
```

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```
ii) #include <stdio.h>
#include <stdlib.h>
int main() {
    char sentence[1000];
    FILE *fptr = fopen("file2.txt", "w");

    if (fptr == NULL) {
        printf("Error");
        exit(1);
    }

    printf("Enter a sentence:\n");
    fgets(sentence, sizeof(sentence), stdin);
    fprintf(fptr, "%s", sentence);
    fclose(fptr);
    return 0;
}
```

```
iv) #include <stdio.h>
#include <unistd.h>
#include <errno.h>
int main() {
    const char *originalFile = "original.txt";
    const char *linkedFile = "linked.txt";
    if (link(originalFile, linkedFile) == 0) {
        printf("Successfully created a hard link: %s -> %s\n", linkedFile, originalFile);
    } else {
        perror("Error creating link");
    }
}
```

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```
if (unlink(linkedFile) == 0) {  
    printf("Successfully unlinked the file %s\n",  
        linkedFile);
```

```
} else {
```

```
    perror("Error unlinking file");
```

```
}
```

```
return 0;
```

```
}
```

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```
V.) #include <stdio.h>
#include <stdlib.h>
int main()
{
    FILE* fptr1, fptr2;
    char filename[100];
    int c;
    printf("Enter the filename to open for reading:");
    scanf("%s", filename);
    fptr1 = fopen(filename, "r");
    if (fptr1 == NULL) {
        printf("Cannot open file %s\n", filename);
        exit(1);
    }
    printf("Enter the filename to open for writing:");
    scanf("%s", filename);
    if
    fptr2 = fopen(filename, "w");
    if (fptr2 == NULL) {
        printf("Cannot open file %s\n", filename);
        exit(1);
    }
    while ( (c = fgetc(fptr1)) != EOF ) {
        fputc(c, fptr2);
    }
    printf("Contents Copied to %s\n", filename);
    fclose(fptr1);
    fclose(fptr2);
    return 0;
}
```

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```
vi) #include <stdio.h>
#include <stdlib.h>
```

```
int main() {
```

```
    const char* filename = "input.txt";
```

```
    FILE* file = fopen(filename, "r");
```

```
    if (file == NULL) {
```

```
        perror("Error opening file");
```

```
        return EXIT_FAILURE;
```

```
    }
```

```
    fseek(file, 0, SEEK_END);
```

```
    long fileSize = ftell(file);
```

```
    if (fileSize == -1) {
```

```
        perror("Error determining file size");
```

```
        fclose(file);
```

```
        return EXIT_FAILURE;
```

```
    }
```

```
    printf("Contents of the file in reverse order:\n");
```

```
    for (long i = fileSize - 1; i >= 0; i--) {
```

```
        fseek(file, i, SEEK_SET);
```

```
        char ch = fgetc(file);
```

```
        if (ch != EOF) {
```

```
            putchar(ch);
```

```
        }
```

```
    }
```

```
    printf("\n");
```

```
    fclose(file);
```

```
    return EXIT_SUCCESS;
```

```
}
```

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2)

```
#include <stdio.h>
```

```
int main() {
```

```
    char input[16];
```

```
    printf("Enter upto 15 characters: ");
```

```
    if (fgets(input, sizeof(input), stdin) != NULL) {
```

```
        char *newline = strchr(input, '\n');
```

```
        if (newline) {
```

```
            *newline = '\0';
```

```
        }
```

```
        printf("You entered: %s\n", input);
```

```
    } else {
```

```
        printf("Error reading input.\n");
```

```
    }
```

```
    return 0;
```

```
}
```

3)

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    FILE *file1, *file2;
```

```
    char buffer[2];
```

```
    size_t bytesRead;
```

```
    file1 = fopen("File1.txt", "r");
```

```
    if (file1 == NULL) {
```

```
        perror("Error opening File1.txt");
```

```
        return EXIT_FAILURE;
```

```
    }
```

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```
file2 = fopen("file2.txt", "a");
```

```
if(file2 == NULL) {
```

```
    perror("Error opening file2.txt");
```

```
    fclose(file1);
```

```
    return EXIT_FAILURE;
```

```
}
```

```
bytesRead = fread(buffer, sizeof(char), 20, file1);
```

```
buffer[bytesRead] = '\0';
```

```
if(fwrite(buffer, sizeof(char), bytesRead, file2) !=  
    bytesRead) {
```

```
    perror("Error writing to file2.txt");
```

```
    fclose(file1);
```

```
    fclose(file2);
```

```
    return EXIT_FAILURE;
```

```
}
```

```
fclose(file1);
```

```
fclose(file2);
```

```
return EXIT_SUCCESS;
```

```
}
```

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4)

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <dirent.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#define BUFFER_SIZE 1024
```

```
int main() {
```

```
    DIR *dir;
```

```
    struct dirent *entry;
```

```
    int mergedFileFD;
```

```
    FILE *inputFile;
```

```
    char buffer[BUFFER_SIZE];
```

```
    size_t bytesRead;
```

```
    dir = opendir(".");
```

```
    if (dir == NULL) {
```

```
        perror("Error opening directory");
```

```
        return EXIT_FAILURE;
```

```
    }
```

```
    mergedFileFD = open("merge.txt",
```

```
        O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```
    if (mergedFileFD == -1) {
```

```
        perror("Error creating merged file");
```

```
        closedir(dir);
```

```
        return EXIT_FAILURE;
```

```
    }
```

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```
while (Centry = readdir(dir)) != NULL {
```

```
    if (strstr(Centry->d_name, ".txt") != NULL) {
```

```
        && strcmp(Centry->d_name, "merged.txt") != 0)
```

```
    {
```

```
        printf("Merging file: %s\n", Centry->d_name);
```

```
        inputFile = fopen(Centry->d_name, "r");
```

```
        if (inputFile == NULL) {
```

```
            perror("Error opening input file");  
            continue;
```

```
        }
```

```
        while ( (bytesRead = fread(buffer, sizeof(char),  
            BUFFER_SIZE, inputFile)) > 0 ) {
```

```
            if (write(mergedFileFD, buffer,  
                bytesRead) == -1) {
```

```
                perror("Error");
```

```
                close(mergedFileFD);
```

```
                closedir(dir);
```

```
                return EXIT_FAILURE;
```

```
            }
```

```
        }
```

```
        fclose(inputFile);
```

```
    }
```

```
}
```

```
closedir(dir);
```

```
return mergedFileFD;
```

```
}
```

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5)

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
```

```
int main() {
```

```
    const char *filename = "fileName.txt";
```

```
    int fd;
```

```
    char buffer[6];
```

```
    off_t fileSize;
```

```
    fd = open(filename, O_RDONLY);
```

```
    if (fd == -1) {
```

```
        perror("Error opening file");
```

```
        return EXIT_FAILURE;
```

```
    }
```

```
    fileSize = lseek(fd, 0, SEEK_END);
```

```
    if (fileSize == -1) {
```

```
        perror("Error determining file size");
```

```
        close(fd);
```

```
        return EXIT_FAILURE;
```

```
    }
```

```
    if (fileSize < 5) {
```

```
        printf("File has less than 5 characters");
```

```
    }
```

```
    if (read(fd, buffer, 5) == -1) {
```

```
        perror("Error reading file");
```

```
        close(fd);
```

```
        return EXIT_FAILURE;
```

```
    }
```

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```
buffer[5] = '\0';  
printf("Last 5 characters: %s\n", buffer);  
close(fd);  
return EXIT_SUCCESS;
```

}

6)

```
#include <stdio.h>  
#include <stdlib.h>  
#include <fcntl.h>  
#include <unistd.h>  
#define BUFFER_SIZE 6  
  
int main() {  
    const char *inputFile = "file1.txt";  
    const char *outputFile = "file2.txt";  
    int inputFD, outputFD;  
    char buffer[BUFFER_SIZE];  
    ssize_t bytesRead, bytesWritten;  
    inputFD = open(inputFile, O_RDONLY);  
    if(inputFD == -1) {  
        perror("Error opening input file");  
        return EXIT_FAILURE;  
    }  
    outputFD = open(outputFile, O_WRONLY |  
        O_CREAT | O_APPEND, 0644);  
    if(outputFD == -1) {  
        perror("Error opening output file");  
        close(inputFD);  
        return EXIT_FAILURE;  
    }
```

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```
if (lseek(inputFD, 14, SEEK_SET) == -1) {  
    perror("Error seeking in input file");  
    close(inputFD);  
    close(outputFD);  
    return EXIT_FAILURE;  
}
```

```
bytesRead = read(inputFD, buffer, BUFFER_SIZE-1);
```

```
if (bytesRead == -1) {  
    perror("Error reading from input file");  
    close(inputFD);  
    close(outputFD);  
    return EXIT_FAILURE;  
}
```

```
buffer[bytesRead] = "\0";  
bytesWritten = write(outputFD, buffer, bytesRead);
```

```
close(inputFD);  
close(outputFD);  
return EXIT_SUCCESS;
```

```
}
```